PHSX 661, 761 Problem Set 1

Due by Tue. Feb. 7th at 11 AM

Required Concepts:

- Natural units
- Fine structure constant, α
- Cross-section and luminosity
- Four-vectors
- Relativistic invariants
- Lorentz transformations
- Center-of-momentum (CM) frame and production/decay angle in CM frame
- Fixed-target and collider system
- Two-body decay kinematics
- Feynman diagrams
- Mandelstam variables

Problem A. The pure QED (photon exchange only) total cross-section for $e^+e^- \to \mu^+\mu^-$ with unpolarized electrons and positrons for $s \gg 4m_\mu^2$ is

$$\sigma = \frac{4\pi\alpha^2}{3s} \ . \tag{1}$$

This expression is in natural units where $\hbar = 1$, $c = 1$. α is the fine structure constant, a dimensionless number related to the electron charge, with a value of $\frac{1}{137.036}$ at $Q^2 = 0$. With natural units we measure momentum and mass in energy units, and distances and times in inverse units of energy. This makes lots of equations much simpler. But we need to know how to put back those missing factors of $\hbar$ and c when actually calculating experimentally observable things like cross-sections and lifetimes. Of course in SI units, $\hbar = 1.055 \times 10^{-34}$ J s, $c = 2.998 \times 10^8$m/s. Or maybe more conveniently for us, $\hbar = 6.58 \times 10^{-22}$ MeV s, and $c = 29.98$ cm/ns.

1. Write down the dimensions of $\hbar$ and c in terms of powers of mass, length, and time.

2. Now include the missing $\hbar$ and c factors in equation 1 based on dimensional arguments, bearing in mind that s, the Lorentz invariant quantity, has dimensions of mass squared.

3. Evaluate this cross-section in units of m^2 and barns for a center-of-momentum mass of $\sqrt{s} = 20$ GeV$/c^2$. (1 barn $= 10^{-24}$ cm^2 $= 10^{-28}$ m^2)

4. An experiment takes data for 10^7 s at a mean instantaneous luminosity of 4×10^{30}cm^{-2}s^{-1} at $\sqrt{s} = 20$ GeV$/c^2$. How many $\mathrm{e^+e^-} \to \mu^+\mu^-$ events would be produced on average?

5. There are a number of conversion constants that are helpful. Calculate $\hbar c$ in units of MeV fm, and calculate $(\hbar c)^2$ in units of GeV^2 mbarn.

Problem B. Particle A (energy E) collides with particle B (at rest), producing particles $C_1, C_2, ..., C_n$. Define $M = m_1 + m_2 + ... + m_n$ and calculate the threshold energy, (ie. minimum value of E), for this reaction in terms of the various particle masses.

Use this result to find the threshold energies for the following reactions assuming the target proton or electron is stationary (fixed-target configuration) and using the Particle Data Group tables to find any needed mass values.

1. $\mathrm{p + p \to p + p} + \pi^0$ (normally just written $\mathrm{p\,p \to p\,p}\,\pi^0$)

2. $\mathrm{p\,p \to p\,p}\; J/\psi$

3. $\mathrm{e^+e^-} \to \mu^+\mu^-$

4. $\mathrm{e^+e^-} \to \tau^+\tau^-$

5. $\mathrm{e^+e^-} \to \mathrm{Z}$

Problem C. Ultra-high energy protons of distant cosmic origin are presumed to be unobservable because of interactions with the 2.7K cosmic microwave background (CMB) radiation. This kinematic limit is known as the GZK (Greisen, Zatsepin, Kuzmin) cut-off. Calculate the proton energy corresponding to the minimum energy required for the reaction: $\mathrm{p}\gamma \to \mathrm{e^+e^-p}$ assuming that the CMB photons have an energy of 6×10^{-4} eV.

Problem D. An $\mathrm{e^+e^-}$ collider, collides 125 GeV electrons with 125 GeV positrons. The collision results in the production of a single photon with an energy of 100 GeV accompanied by some undetected particles. Calculate the mass of the undetected particle system using four-momentum conservation (you may neglect the electron and positron masses in your calculation). The quantity you are calculating is termed the missing mass, or sometimes, the recoil mass to the visible system. So in this case the recoil mass to the photon.

Problem E. Calculate the invariant mass of a 3-particle system consisting of one charged pion and two photons. The components of the charged pion's 3-momentum are: (0.15, 0.60, 0.40) GeV. The components of the first photon's 3-momentum are: (0.20, -0.50, 0.20) GeV. The components of the second photon's 3-momentum are: (-0.10, 0.40, 0.90) GeV. You should be able to look up the charged pion mass (and even the photon mass ...) in the PDG.

Problem F. The π^0 meson mostly (98.8% probability) decays into two photons. The π^0 mass, M, is 135.0 MeV (using natural units – get used to it ...), and of course the photons are massless.

- Show that the following expression holds between the decay photon energies and the photon-photon separation angle in all reference frames,

$$m_{\pi^0}^2 = 2E_{\gamma_1}E_{\gamma_2}(1 - \cos\psi_{\gamma_1\gamma_2})$$

  (Hint: there is no need to do an explicit Lorentz transformation).

- Calculate the minimum opening angle between the photons in the laboratory frame for a π^0 with an energy of 100 GeV.

- Evaluate the energies of the two photons in the lab frame if one of the photons has a center-of-momentum production angle of 45° with respect to the boost direction. (Hint: draw a picture of the decay in the rest frame of the π^0 and perform a Lorentz transformation (ie. boost) of the two photon 4-vectors into the lab frame.

Problem G. Two-body decay, one massless. The charged pion of mass, M, decays by the weak interaction with a mean life of $(2.6033 \pm 0.0005) \times 10^{-8}$s. Its main decay mode is

$$\pi^+ \to \mu^+\nu_\mu \ .$$

(it is an interesting issue why this mode dominates over the more kinematically favored $\pi^+ \to \mathrm{e}^+\nu_\mathrm{e}$). Find the center-of-mass momentum of the muon with mass, m, assuming that the neutrino mass is zero.

Problem H. Two-body decay, both decay products with the same mass. The short-lived version of the neutral kaon, K^0_S, of mass, M, can decay to two charged pions, each of mass, m,

$$\mathrm{K}^0_\mathrm{S} \to \pi^+\pi^- \ .$$

1. Find an expression for the momentum of the pion in the rest frame of the K^0_S.

2. Find expressions for the components of the pion lab momenta, parallel and perpendicular to the boost direction for various values of the center-of-mass decay angle, θ^*, for a K^0_S traveling with lab frame momentum of 10 GeV. (Hint: you should figure out what value of β this corresponds to).

Problem I. Two-body decay, decay products with different mass. The excited charged D meson (mass M) can decay to the neutral D meson (mass m_1) accompanied by a "slow (charged) pion" of mass m_2,

$$\mathrm{D}^{*+} \to \mathrm{D}^0\pi^+ \ .$$

1. Find an expression for the momentum of the pion in the rest frame of the D^{*+}.

2. Find an expression for the magnitude of the pion momentum in the lab for various center-of-mass decay angles for a D^{*+} traveling with lab frame momentum of 20 GeV. (Hint: again you should figure out what value of β this corresponds to).

Problem J.
Draw the 3 lowest order Feynman diagrams consisting of 2 vertices for the process $u\overline{u} \to W^+W^-$.

Problem K.
Draw the 5 lowest order Feynman diagrams consisting of 3 vertices for the process $e^+e^- \to \nu_e\overline{\nu}_e\gamma$. (If you only manage to find three, note that it matters which particle emits the gauge boson.)

Problem L.
Draw the 10 lowest order Feynman diagrams consisting of 4 vertices for the process $e^+e^- \to \overline{u}d\mu^+\nu_\mu$.

Problem M. For the process $1 + 2 \to 3 + 4$ with the conventional definition of the Mandelstam variables (s, t, u). Show that

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2$$